

Symmetric polynomials and modules over affine $\mathfrak{sl}(2)$ at admissible levels

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Joint work with David Ridout

Conference on Lie algebras, vertex operator
algebras, and related topics

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Affine vertex operator algebras

Let $\mathfrak{g}$ be a simple Lie algebra with affinisation $\widehat{\mathfrak{g}} = \mathfrak{g} \otimes \mathbb{C}[t, t^{-1}] \oplus \mathbb{C}K$.
Then, for $k \neq -h^\vee$,

$$V_k(\mathfrak{g}) = \text{Ind}_{\mathfrak{g} \otimes \mathbb{C}[t] \oplus \mathbb{C}K}^{\widehat{\mathfrak{g}}} \mathbb{C}_k, \quad K|_{\mathbb{C}_k} = k \cdot \text{id}, \quad \mathfrak{g} \otimes \mathbb{C}[t]|_{\mathbb{C}_k} = 0,$$

is a *universal affine vertex operator algebra*.

For certain levels k , there exist proper ideals.

$$L_k(\mathfrak{g}) = \frac{V_k(\mathfrak{g})}{\langle \text{max ideal} \rangle}.$$

Idea and goal

Determine module theory of $L_k(\mathfrak{g})$ from that of $V_k(\mathfrak{g})$.

Example $\mathfrak{g} = \mathfrak{sl}(2) = \text{span}\{E, H, F\}$

For $\mathfrak{g} = \mathfrak{sl}(2)$, there exists a (unique) proper ideal I if and only if

$$k+2 = \frac{u}{v}, \quad u \geq 2, v \geq 1, \gcd(u, v) = 1,$$

$$L_k(\mathfrak{sl}(2)) = \frac{V_k(\mathfrak{sl}(2))}{I}.$$

Such levels are called **admissible**. The ideal is generated by a singular vector χ of $\mathfrak{sl}(2)$ -weight $2(u-1)$ and conformal weight $(u-1)v$.

Integral levels

For $v = 1$,

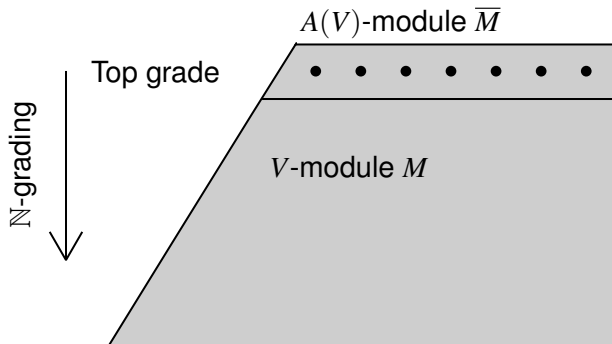
$$\chi = (E_{-1})^{u-1} 1_{u-2}.$$

The Zhu algebra

Moral “definition”: Zhu algebra of vertex operator algebra V

$$A(V) \simeq \{0\text{-modes of } V \text{ acting on vectors annihilated by pos. modes.}\}$$

There is a 1-1 correspondence between simple $\mathbb{N}$ -gradable modules over a vertex operator algebra V and simple modules over the Zhu algebra $A(V)$.



Classification strategy

Let $\pi : V \twoheadrightarrow A(V)$.

Theorem [Frenkel, Zhu]

- For $V_k(\mathfrak{g})$, Zhu's algebra is $A(V_k(\mathfrak{g})) \simeq U(\mathfrak{g})$.
- For any ideal $I \subset V_k(\mathfrak{g})$, the image $\pi(I)$ is an ideal of $A(V_k(\mathfrak{g}))$ and $A\left(\frac{V_k(\mathfrak{g})}{I}\right) = \frac{A(V_k(\mathfrak{g}))}{\pi(I)}$
- For $\chi \in V_k(\mathfrak{g})$ singular, such that $\langle \chi \rangle = I \Rightarrow \langle \pi(\chi) \rangle = \pi(I)$.

Classifying $\mathbb{N}$ -gradable weight $L_k(\mathfrak{g})$ -modules

- A $V_k(\mathfrak{g})$ -module M is an $L_k(\mathfrak{g})$ -module $\iff I$ annihilates M .
- Simple $\mathbb{N}$ -gradable $L_k(\mathfrak{g})$ -modules $\xleftrightarrow{1-1}$ simple $\frac{U(\mathfrak{g})}{\pi(I)}$ -weight modules.
- $U(\mathfrak{g})$ -weight modules $\Rightarrow \frac{U(\mathfrak{g})}{\pi(I)}$ -weight modules $\Rightarrow \mathbb{N}$ -gradable $L_k(\mathfrak{g})$ -modules.

Example $\mathfrak{g} = \mathfrak{sl}(2) = \text{span}\{E, H, F\}$

Theorem [Gabriel]

Any simple $\mathfrak{sl}(2)$ weight module with finite-dimensional weight spaces is isomorphic to one of the following:

- Finite-dimensional modules $\mathcal{F}_\lambda$, $\lambda \in \mathbb{Z}_{\geq 0}$. Highest and lowest weight. Weights: $\lambda, \lambda - 2, \dots, 2 - \lambda, -\lambda$
- Infinite-dimensional highest weight modules $\mathcal{H}_\lambda$, $\lambda \in \mathbb{C} \setminus \mathbb{Z}_{\geq 0}$. Weights: $\lambda, \lambda - 2, \lambda - 4, \dots$
- Infinite-dimensional lowest weight modules $\mathcal{L}_\lambda$, $\lambda \in \mathbb{C} \setminus \mathbb{Z}_{\leq 0}$. Weights: $\dots, \lambda + 4, \lambda + 2, \lambda$
- Infinite-dimensional weight modules $\mathcal{W}_{\lambda; \Delta}$, $\lambda, \Delta \in \mathbb{C}$ and $2\Delta \neq \mu (\mu + 2)$ for any $\mu \in \lambda + 2\mathbb{Z}$, where Δ is the eigenvalue of the quadratic Casimir and $\mathcal{W}_{\lambda; \Delta} \cong \mathcal{W}_{\lambda+2; \Delta}$. Neither highest nor lowest weight.
Weights: $\dots, 2 + \lambda, \lambda, \lambda - 2, \dots$

All weight spaces are 1-dimensional.

Example $\mathfrak{g} = \mathfrak{sl}(2) = \text{span}\{E, H, F\}$

- For $k \in \mathbb{Z}_{\geq 0}$, the ideal of $V_k(\mathfrak{sl}(2))$ is generated by the singular vector $\chi = (E_{-1})^{k+1} 1_k$.
- In $A(V_k(\mathfrak{sl}(2))) \simeq U(\mathfrak{sl}(2))$, we have $\pi((E_{-1})^{k+1} 1_k) = E^{k+1}$.
- The generator E is nilpotent in $A\left(\frac{V_k(\mathfrak{sl}(2))}{\langle (E_{-1})^{k+1} 1_k \rangle}\right) \simeq \frac{U(\mathfrak{sl}(2))}{\langle E^{k+1} \rangle}$.
- The simple $\mathbb{N}$ -gradable $L_k(\mathfrak{sl}(2))$ -modules are those with top grade $\mathcal{F}_\lambda$, $\lambda = 0, \dots, k$.

Upshot

Easy if the singular vector is easy, very hard if not.

General admissible levels

Let

$$k+2 = \frac{u}{v}, \quad \lambda_{r,s} = r-1 - s\frac{u}{v}, \quad \Delta_{r,s} = \frac{r^2-1}{2} + \frac{s^2}{2} \frac{u^2}{v^2} - rs\frac{u}{v}.$$

Theorem [Adamović, Milas] [Ridout, SW]

Any simple $\mathbb{N}$ -gradable $L_k(\mathfrak{sl}(2))$ -module is isomorphic to one of the following:

- The simple quotients induced from the finite-dimensional modules $\mathcal{F}_{r-1}$, where $1 \leq r \leq u-1$.
- The simple quotients induced from the infinite-dimensional highest weight modules $\mathcal{H}_{\lambda_{r,s}}$, where $1 \leq r \leq u-1$ and $1 \leq s \leq v-1$.
- The simple quotients induced from the infinite-dimensional lowest weight modules $\mathcal{L}_{-\lambda_{r,s}}$, where $1 \leq r \leq u-1$ and $1 \leq s \leq v-1$.
- The simple quotients induced from the infinite-dimensional weight modules $\mathcal{W}_{\lambda;\Delta_{r,s}}$, where $1 \leq r \leq u-1$ and $1 \leq s \leq v-1$, $2\Delta_{r,s} \neq \mu(\mu+2)$ for all $\mu \in \lambda + 2\mathbb{Z}$ and $\mathcal{W}_{\lambda;\Delta_{r,s}} \cong \mathcal{W}_{\lambda;\Delta_{u-r,v-s}}$.

Proof idea: Wakimoto free field realisation

$V_k(\mathfrak{sl}(2))$ is a vertex operator subalgebra of $\{\text{rank 1 Heisenberg}\} \otimes \{\beta\gamma\text{-ghosts}\}$.

$$a(z)a(w) \sim \frac{1}{(z-w)^2}, \quad \gamma(z)\beta(w) \sim \frac{1}{z-w}, \quad \beta(z)\beta(w) \sim 0 \sim \gamma(z)\gamma(w).$$

$$E(z) = \beta(z),$$

$$H(z) = 2 : \beta(z)\gamma(z) : + \sqrt{2k+4}a(z),$$

$$F(z) = : \beta(z)\gamma(z)\gamma(z) : + \sqrt{2k+4} : a(z)\gamma(z) : + k\partial\gamma(z).$$

Screening operator

$$S(z) = : \beta(z) \exp \left(-\sqrt{\frac{2}{k+2}} \phi(z) \right) :, \quad \partial\phi(z) = a(z).$$

Proof idea: Wakimoto free field realisation

The singular vector of $V_{\frac{u}{v}-2}(\mathfrak{sl}(2))$, in the free field realisation, can be realised by the screening operator.

$$\begin{aligned} S^{[u-1]}|q\rangle &= \int S(z_1) \cdots S(z_{u-1}) |q\rangle dz \\ &= \int \beta(z_1) \cdots \beta(z_{u-1}) \\ &\times \prod_{1 \leq i \neq j \leq u-1} \left(1 - \frac{z_i}{z_j}\right)^{\frac{v}{u}} \prod_{i=1}^{u-1} z_i^{-v-1} \prod_{m \geq 1} \exp\left(-\sqrt{\frac{2v}{u}} \frac{\mathfrak{p}_m(z) a_{-m}}{m}\right) |q\rangle \frac{dz_1 \cdots dz_{u-1}}{z_1 \cdots z_{u-1}}, \end{aligned}$$

where

$$q = (u-1)\sqrt{\frac{2v}{u}}, \quad \mathfrak{p}_m(z) = \sum_{i=1}^{u-1} z_i^m.$$

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 &\times \underbrace{\prod_{1 \leq i \neq j \leq u-1} \left(1 - \frac{z_i}{z_j}\right)^{\frac{v}{u}}}_{\text{Inner prod. of Jack symm. poly.}} \prod_{i=1}^{u-1} z_i^{-v-1} \prod_{m \geq 1} \exp\left(-\sqrt{\frac{2v}{u}} \frac{\mathbf{p}_m(z) a_{-m}}{m}\right) |q\rangle \frac{dz_1 \cdots dz_{u-1}}{z_1 \cdots z_{u-1}},
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 \end{aligned}$$

Proof idea: The generator of the ideal

Choose a generator of $\mathfrak{sl}(2)$ -weight 0.

$$\chi = F_0^{u-1} S^{[u-1]} |q\rangle = \text{const} \cdot S^{[u-1]} \gamma_0^{u-1} |q\rangle$$

Compute eigenvalue of zero-mode χ_0 on a general top grade vector to determine image in $A(V_{\frac{u}{v}-2}(\mathfrak{sl}(2)))$.

$$\chi_0 |p, \tau\rangle = f(p, \tau) |p, \tau\rangle, \quad p = \text{Heisenberg weight}, \quad \tau = \beta\gamma\text{-weight}.$$

Theorem [Ridout, SW]

- 1 The polynomial $f(p, \tau)$, in free field data, is also a polynomial in $\mathfrak{sl}(2)$ -data.

- 2
$$f(\lambda, \Delta) = g_u(\lambda, \Delta) \prod_{r,s} (\Delta - \Delta_{r,s}),$$

$$g_{u+2}(\lambda, \Delta) = \frac{(2u+1)\lambda}{(u+1)^2} g_{u+1}(\lambda, \Delta) - \frac{2\Delta - (u-1)(u+1)}{(u+1)^2} g_u(\lambda, \Delta)$$

$$g_1(\lambda, \Delta) = 1, \quad g_2(\lambda, \Delta) = \lambda.$$

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$$A(L_k(\mathfrak{sl}(2))) = \frac{U(\mathfrak{sl}(2))}{\langle f(H, Q) \rangle}$$

Outlook

Computing the Zhu algebra becomes almost algorithmic when using free fields and symmetric polynomials. Successfully applied to:

- Virasoro minimal models
- $\mathfrak{sl}(2)$ minimal models
- $W_{p,q}$ -triplet models

Work in progress:

- $N = 1$ Virasoro minimal models
- $\mathfrak{osp}(1|2)$ minimal models

Future plans:

- higher rank affine Kac-Moody superalgebras
- higher rank W -algebras

The End

Thank you!